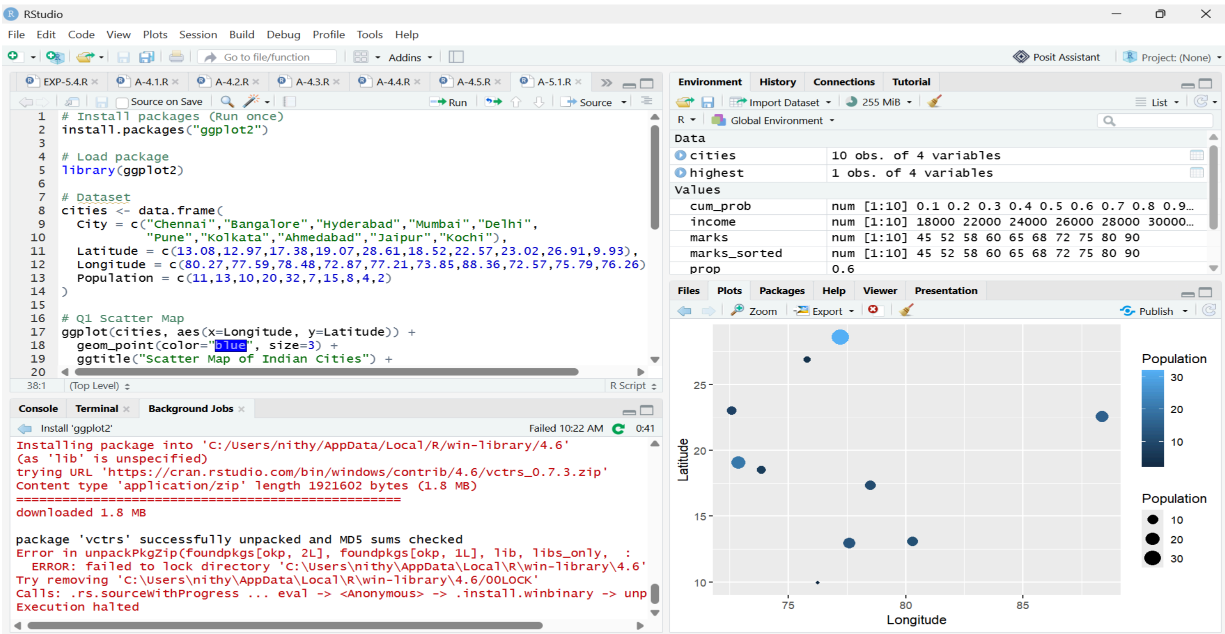
QUESTION-1:



CODE:

```
# Install packages (Run once)
install.packages("ggplot2")

# Load package
library(ggplot2)

# Dataset
cities <- data.frame(
  City = c("Chennai","Bangalore","Hyderabad","Mumbai","Delhi",
           "Pune","Kolkata","Ahmedabad","Jaipur","Kochi"),
  Latitude = c(13.08,12.97,17.38,19.07,28.61,18.52,22.57,23.02,26.91,9.93),
  Longitude = c(80.27,77.59,78.48,72.87,77.21,73.85,88.36,72.57,75.79,76.26),
  Population = c(11,13,10,20,32,7,15,8,4,2)
)

# Q1 Scatter Map
ggplot(cities, aes(x=Longitude, y=Latitude)) +
  geom_point(color="blue", size=3) +
  ggtitle("Scatter Map of Indian Cities") +
  xlab("Longitude") +
  ylab("Latitude")

# Q2 Bubble Map
ggplot(cities, aes(Longitude, Latitude, size=Population)) +
```

```
  geom_point(color="red", alpha=0.7) +
  ggtitle("Bubble Map Based on Population")

# Q3 Labels
ggplot(cities, aes(Longitude, Latitude, size=Population)) +
  geom_point(color="blue") +
  geom_text(aes(label=City), vjust=-1)

# Q4 Colors Based on Population
ggplot(cities, aes(Longitude, Latitude,
                   size=Population,
                   color=Population)) +
  geom_point()

# Q5 Highest Population
highest <- cities[which.max(cities$Population),]
print(highest)
```

Q1: Scatter map created using latitude and longitude.

Q2: Bubble map created where bubble size represents population.

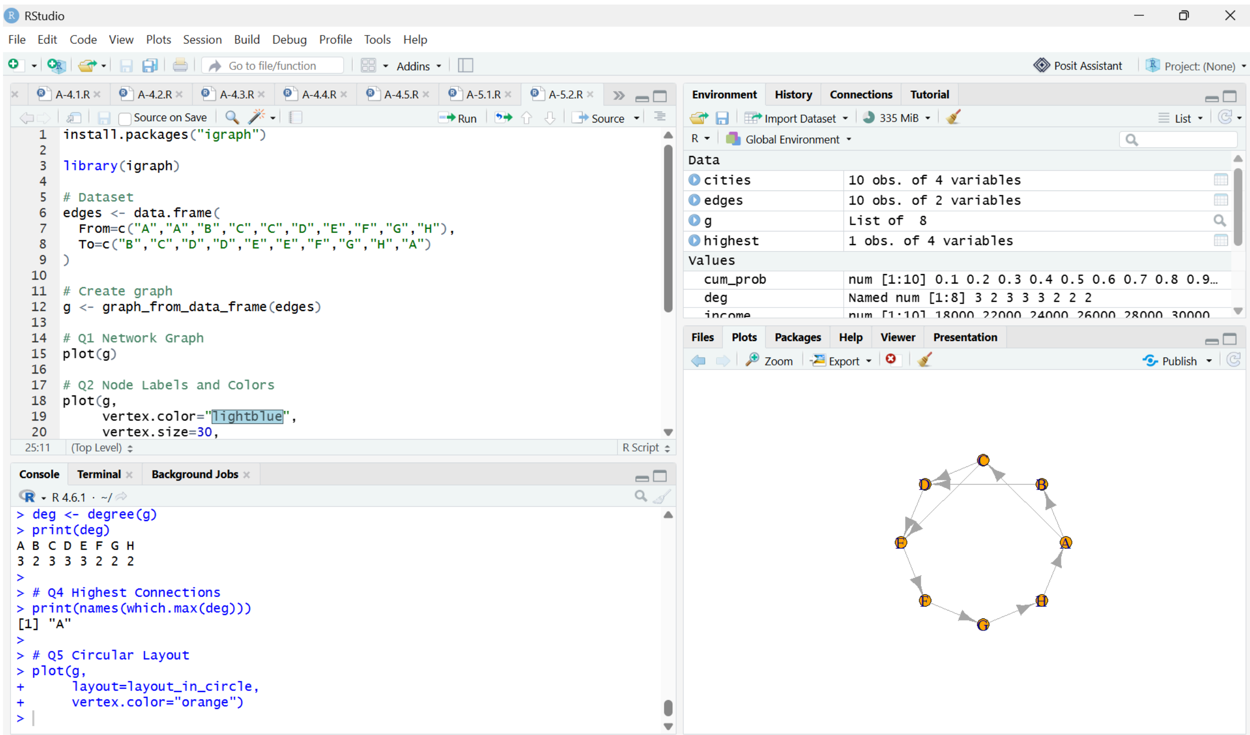
Q3: City labels displayed using `geom_text()`.

Q4: Colors represent different population sizes.

Q5 Answer:

Delhi
Population = 32 Million

QUESTION-2:



CODE:

```
install.packages("igraph")

library(igraph)

# Dataset
edges <- data.frame(
  From=c("A","A","B","C","C","D","E","F","G","H"),
  To=c("B","C","D","D","E","E","F","G","H","A")
)

# Create graph
g <- graph_from_data_frame(edges)

# Q1 Network Graph
plot(g)

# Q2 Node Labels and Colors
plot(g,
     vertex.color="lightblue",
     vertex.size=30,
```

```
    vertex.label.cex=1)

# Q3 Degree
deg <- degree(g)
print(deg)

# Q4 Highest Connections
print(names(which.max(deg)))

# Q5 Circular Layout
plot(g,
    layout=layout_in_circle,
    vertex.color="orange")
```

Q1: Network graph created using igraph.

Q2: Labels and node colors displayed.

Q3 Degree Values

A = 3
B = 2
C = 3
D = 3
E = 3
F = 2
G = 2
H = 2

Q4 Answer

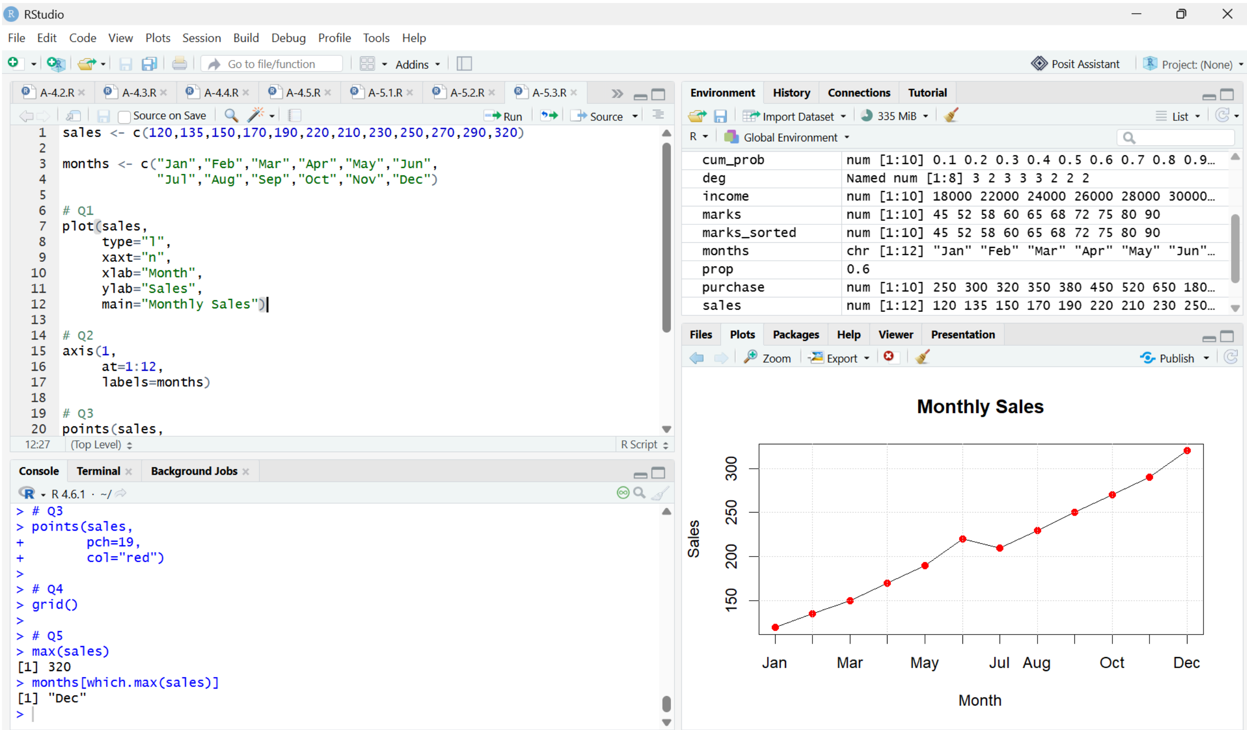
Highest connected nodes are:

A
C
D
E

Each has Degree = **3**

Q5: Circular network plotted.

QUESTION-3:



CODE:

```
sales <- c(120,135,150,170,190,220,210,230,250,270,290,320)

months <- c("Jan","Feb","Mar","Apr","May","Jun",
          "Jul","Aug","Sep","Oct","Nov","Dec")

# Q1
plot(sales,
    type="l",
    xaxt="n",
    xlab="Month",
    ylab="Sales",
    main="Monthly Sales")

# Q2
axis(1,
    at=1:12,
    labels=months)

# Q3
points(sales,
```

```
    pch=19,
    col="red")

# Q4
grid()

# Q5
max(sales)
months[which.max(sales)]
```

Q1: Line chart created.

Q2: Month names added using `axis()`.

Q3: Points highlighted using `pch=19`.

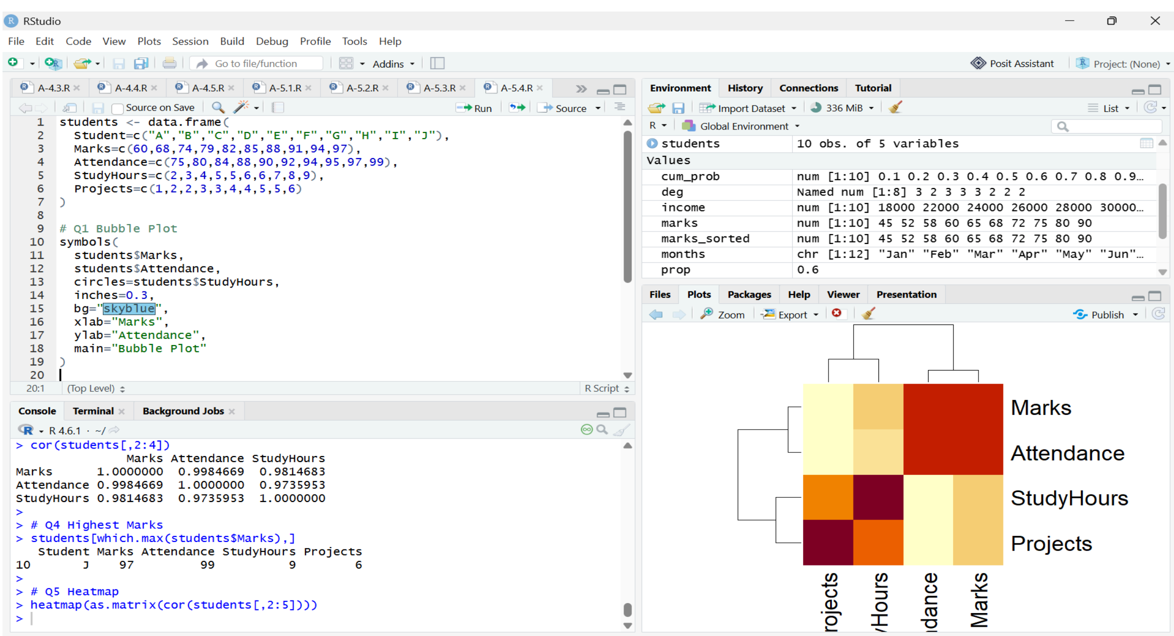
Q4: Grid lines displayed.

Q5 Answer

December

Sales = ₹320 Lakhs

QUESTION-4:



CODE:

```
students <- data.frame(
Student=c("A","B","C","D","E","F","G","H","I","J"),
Marks=c(60,68,74,79,82,85,88,91,94,97),
Attendance=c(75,80,84,88,90,92,94,95,97,99),
StudyHours=c(2,3,4,5,5,6,6,7,8,9),
Projects=c(1,2,2,3,3,4,4,5,5,6)
)

# Q1 Bubble Plot
symbols(
students$Marks,
students$Attendance,
circles=students$StudyHours,
inches=0.3,
bg="skyblue",
xlab="Marks",
ylab="Attendance",
main="Bubble Plot"
)

# Q2 Scatter Plot Matrix
pairs(students[,2:5])

# Q3 Correlation
cor(students[,2:4])

# Q4 Highest Marks
students[which.max(students$Marks),]

# Q5 Heatmap
heatmap(as.matrix(cor(students[,2:5])))
```

Q1: Bubble plot created.

Q2: Scatter plot matrix created.

Q3 Correlation

All variables show **strong positive correlation** (close to **1**).

Q4 Answer

Student J

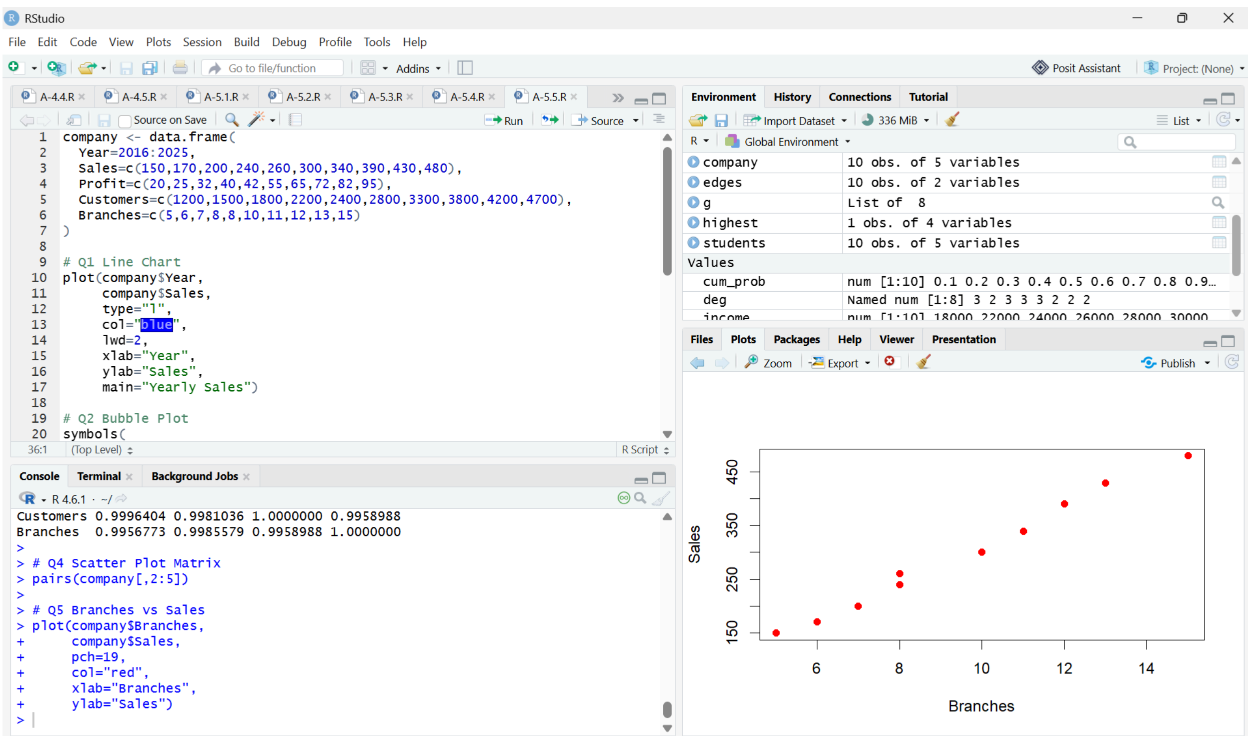
Marks = 97

Study Hours = 9

Student J has the **largest bubble**.

Q5: Heatmap displays strong positive relationships among all numerical variables.

QUESTION-5:



CODE:

```
company <- data.frame(
Year=2016:2025,
Sales=c(150,170,200,240,260,300,340,390,430,480),
Profit=c(20,25,32,40,42,55,65,72,82,95),
Customers=c(1200,1500,1800,2200,2400,2800,3300,3800,4200,4700),
Branches=c(5,6,7,8,8,10,11,12,13,15)
)

# Q1 Line Chart
plot(company$Year,
    company$Sales,
    type="l",
```

```
    col="blue",
    lwd=2,
    xlab="Year",
    ylab="Sales",
    main="Yearly Sales")

# Q2 Bubble Plot
symbols(
company$Sales,
company$Profit,
circles=company$Customers,
inches=0.3,
bg="green",
xlab="Sales",
ylab="Profit",
main="Sales vs Profit"
)

# Q3 Correlation Matrix
cor(company[,2:5])

# Q4 Scatter Plot Matrix
pairs(company[,2:5])

# Q5 Branches vs Sales
plot(company$Branches,
    company$Sales,
    pch=19,
    col="red",
    xlab="Branches",
    ylab="Sales")
```

Q1

Temporal line chart showing yearly sales created successfully.

Q2

Bubble plot created with:

- X-axis = Sales
- Y-axis = Profit
- Bubble size = Customers

Q3

Correlation matrix indicates **very strong positive correlations** among Sales, Profit, Customers, and Branches (values close to **1**).

Q4

Scatter plot matrix shows a clear upward linear relationship among all variables.

Q5 Interpretation

- Sales increase steadily from **₹150 Lakhs (2016)** to **₹480 Lakhs (2025)**.
- Profit also increases from **₹20 Lakhs** to **₹95 Lakhs**.
- The number of branches grows from **5** to **15**.
- Customer count increases from **1,200** to **4,700**.
- The visualizations indicate a **strong positive association** between the number of branches and both sales and profits. As branches expand over the years, sales, profits, and customers also increase, suggesting that business expansion is associated with improved overall performance.